Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

AGILE DEVELOPMENT AND DEVOPS

Lab Fat exam

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COURSENAME: Agile development and devops

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Library Book Mnagement System

Tasks:

1) Develop book listing and lending logic

1A)

Code:

```
import java.util.*;

class Book {
    int id;
    String title;
    boolean isAvailable;

    Book(int id, String title) {
        this.id = id;
        this.title = title;
        this.isAvailable = true;
    }
}

class Library {
    List<Book> books = new ArrayList<>();

    void addBook(int id, String title) {
        books.add(new Book(id, title));
    }

    void listBooks() {
```

```
        for (Book b : books) {
            System.out.println(b.id + " - " + b.title + " - " +
                (b.isAvailable ? "Available" : "Lent"));
        }
    }

    boolean lendBook(int id) {
        for (Book b : books) {
            if (b.id == id && b.isAvailable) {
                b.isAvailable = false;
                return true;
            }
        }
        return false;
    }

    boolean returnBook(int id) {
        for (Book b : books) {
            if (b.id == id && !b.isAvailable) {
                b.isAvailable = true;
                return true;
            }
        }
        return false;
    }
}

public class App {
    public static void main(String[] args) {
        Library lib = new Library();
```

```
        lib.addBook(1, "Java");
        lib.addBook(2, "Python");

        System.out.println("Before Lending:");
        lib.listBooks();

        lib.lendBook(1);

        System.out.println("\nAfter Lending:");
        lib.listBooks();
    }
}
```

2) Write Junit tests for lending success/failure scenarios

2A)

Code:

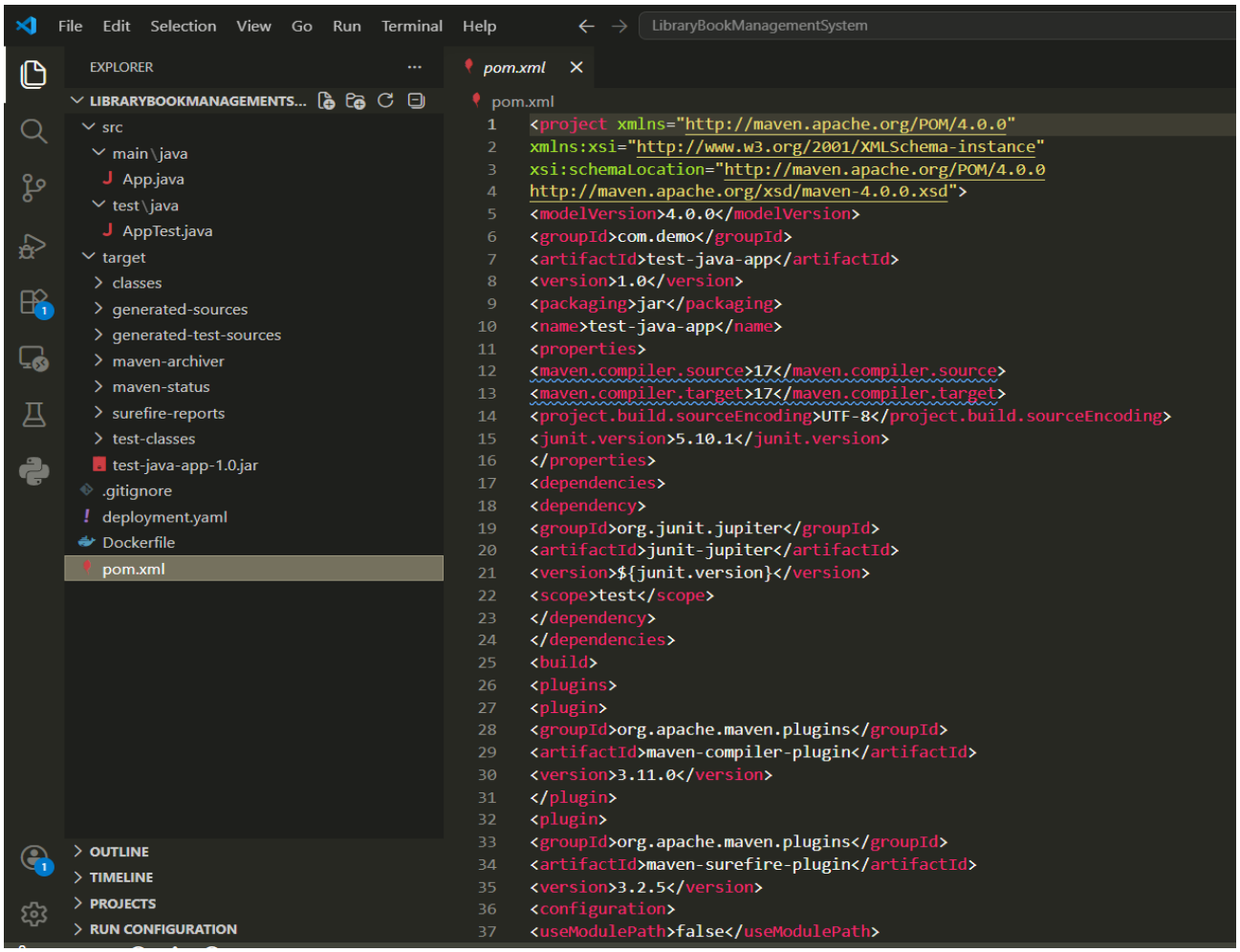
```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class AppTest {

    @Test
    void testLendBookSuccess() {
        Library lib = new Library();
        lib.addBook(1, "Java");

        boolean result = lib.lendBook(1);

        assertTrue(result);
```

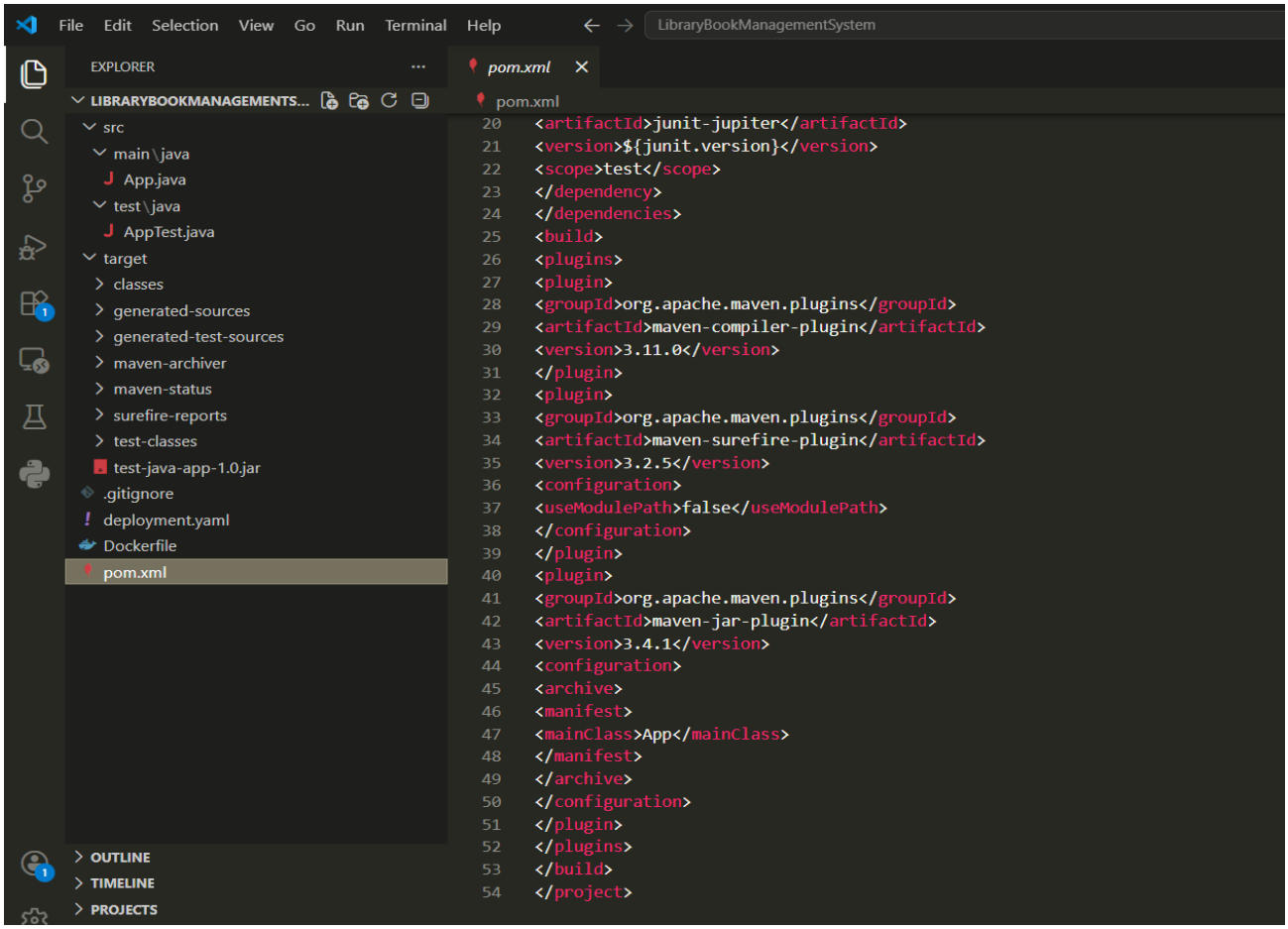
```
    }

    @Test
    void testLendBookFailure() {
        Library lib = new Library();
        lib.addBook(1, "Java");

        lib.lendBook(1); // already lent
        boolean result = lib.lendBook(1);

        assertFalse(result);
    }

    @Test
    void testReturnBook() {
        Library lib = new Library();
        lib.addBook(1, "Java");

        lib.lendBook(1);
        boolean result = lib.returnBook(1);

        assertTrue(result);
    }
}
```

3)Implement CI/CD pipeline with automated testing

3A)

POM.XML:


File Edit Selection View Go Run Terminal Help
LibraryBookManagementSystem
EXPLORER
LIBRARYBOOKMANAGEMENTS...
src
main\java
App.java
test\java
AppTest.java
target
classes
generated-sources
generated-test-sources
maven-archiver
maven-status
surefire-reports
test-classes
test-java-app-1.0.jar
.gitignore
deployment.yaml
Dockerfile
pom.xml
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
pom.xml
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.demo</groupId>
<artifactId>test-java-app</artifactId>
<version>1.0</version>
<packaging>jar</packaging>
<name>test-java-app</name>
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
<junit.version>5.10.1</junit.version>
</properties>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<version>${junit.version}</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>3.2.5</version>
<configuration>
<useModulePath>false</useModulePath>



mvn clean test

```
PS C:\Users\chara\OneDrive\Documents\userregistrationsystem> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\chara\OneDrive\Documents\userregistrationsystem\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\OneDrive\Documents\userregistrationsystem\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\OneDrive\Documents\userregistrationsystem\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.061 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
```

```
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.061 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.644 s
[INFO] Finished at: 2026-04-13T10:00:03+05:30
[INFO] ------------------------------------------------------------------------
```

mvn clean package

```
PS C:\Users\chara\OneDrive\Documents\userregistrationsystem> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\chara\OneDrive\Documents\userregistrationsystem\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\OneDrive\Documents\userregistrationsystem\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\OneDrive\Documents\userregistrationsystem\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
```

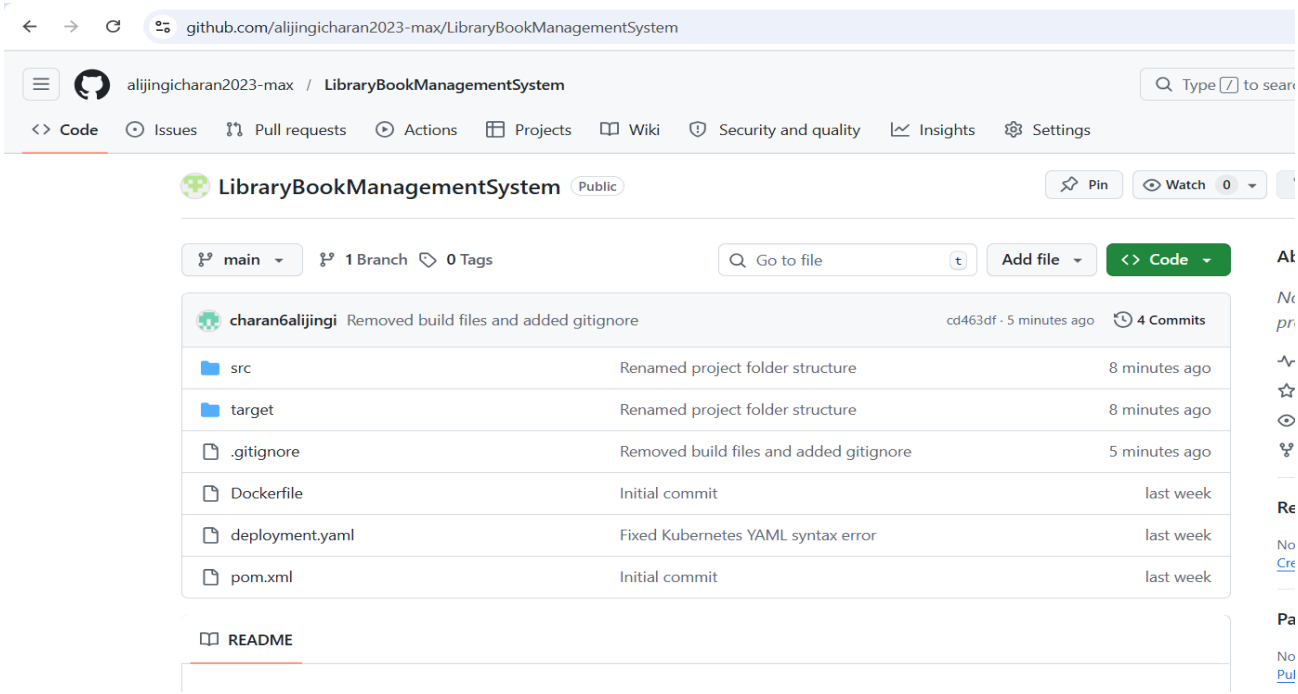
```
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.085 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\chara\OneDrive\Documents\userregistrationsystem\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.475 s
[INFO] Finished at: 2026-04-13T10:07:57+05:30
[INFO] ------------------------------------------------------------------------
```

Target folder created:

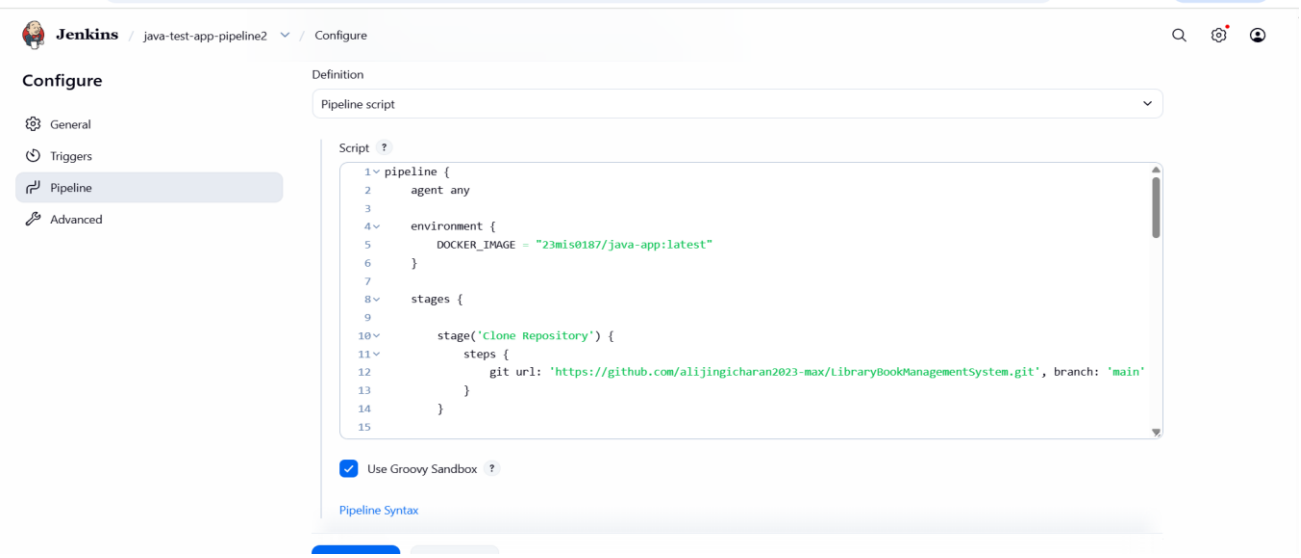
Name	Status	Date modified	Type	Size
src		06-04-2026 10:05 AM	File folder	
target		13-04-2026 10:00 AM	File folder	
deployment.yaml		06-04-2026 10:49 AM	Yaml Source File	2 KB
Dockerfile		06-04-2026 10:25 AM	File	1 KB
pom.xml		06-04-2026 10:12 AM	Microsoft Edge HT...	2 KB

```
PS C:\Users\chara\OneDrive\Documents\LibraryBookManagementSystem> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 295 bytes | 147.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/alijingicharan2023-max/test-java-app.git
```

```
PS C:\Users\chara\OneDrive\Documents\LibraryBookManagementSystem> git remote set-url origin https://github.com/alijingicharan2023-max/LibraryBookManagementS
ystem.git
PS C:\Users\chara\OneDrive\Documents\LibraryBookManagementSystem> git push origin main
Enumerating objects: 64, done.
Counting objects: 100% (64/64), done.
Delta compression using up to 12 threads
Compressing objects: 100% (43/43), done.
Writing objects: 100% (64/64), 15.39 KiB | 656.00 KiB/s, done.
Total 64 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
```



Pipeline script:



Pipeline script

Script

```
        stage('Build with Maven') {
            steps {
                bat 'mvn clean package'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat "docker build -t %DOCKER_IMAGE% ."
            }
        }

        stage('Push to DockerHub') {
            steps {
```

Use Groovy Sandbox

Pipeline Syntax

Script ?

```
            steps {
                withCredentials([
                    usernamePassword(
                        credentialsId: 'dockerhub-creds',
                        usernameVariable: '23mis0187',
                        passwordVariable: 'Suribabu@1234'
                    )
                ]) {
                    bat """
                    echo %DOCKER_PASS% | docker login -u %DOCKER_USER% --password-stdin
                    docker push %DOCKER_IMAGE%
                    """
                }
            }
        }
```

Use Groovy Sandbox ?

Pipeline Syntax

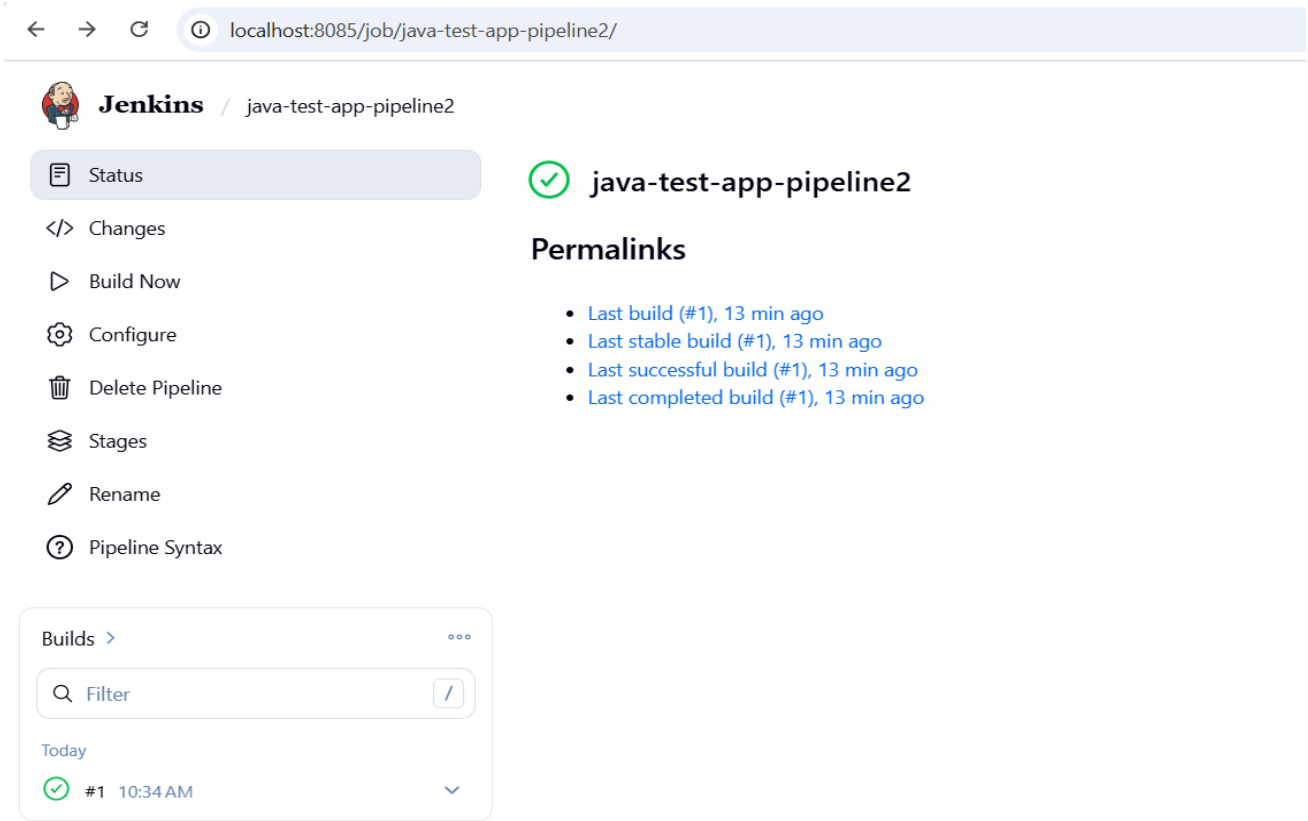
Script ?

```

        stage('Deploy to Kubernetes') {
            steps {
                bat 'kubectl apply -f deployment.yaml'
            }
        }

    }

    post {
        success {
            echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check logs.'
```

Use Groovy Sandbox ?

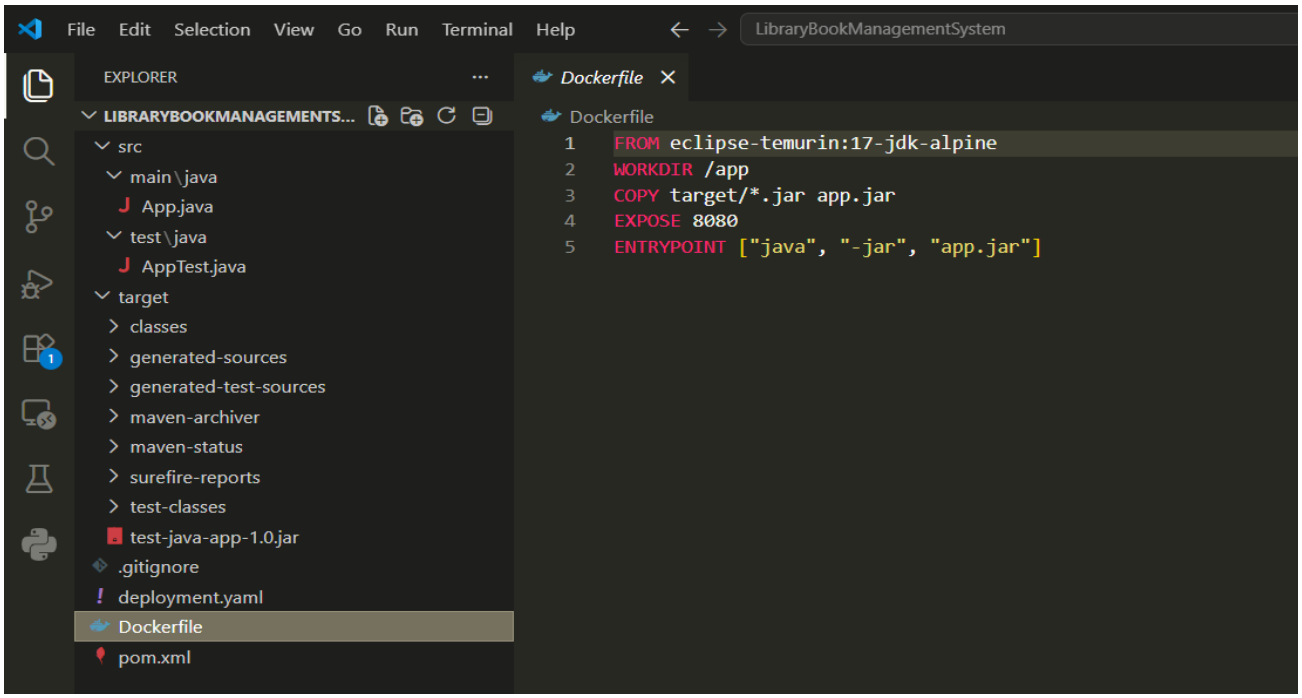
Script ?

```
                bat 'kubectl apply -f deployment.yaml'
            }
        }

    }

    post {
        success {
            echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check logs.'
        }
    }
}
```

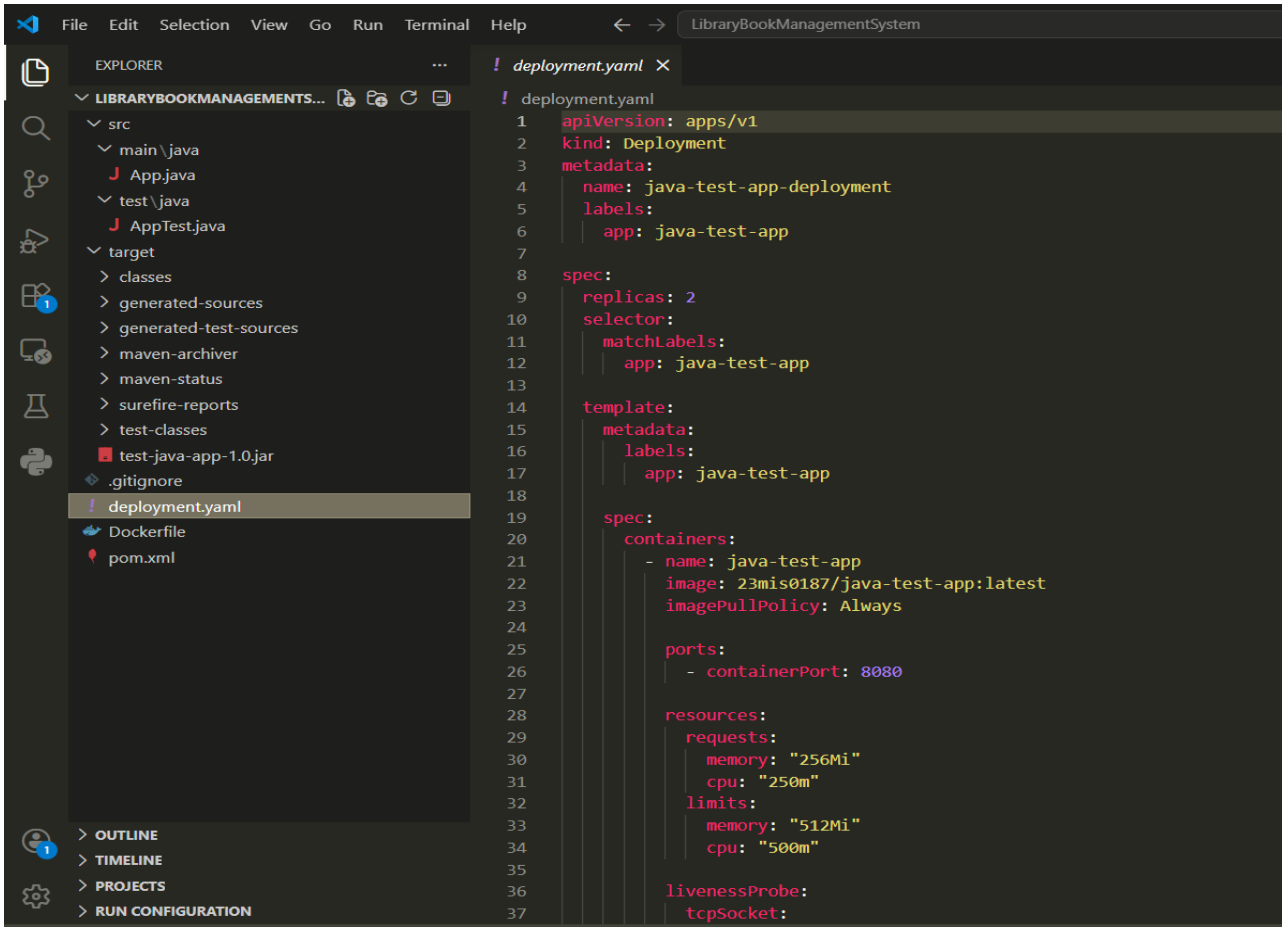


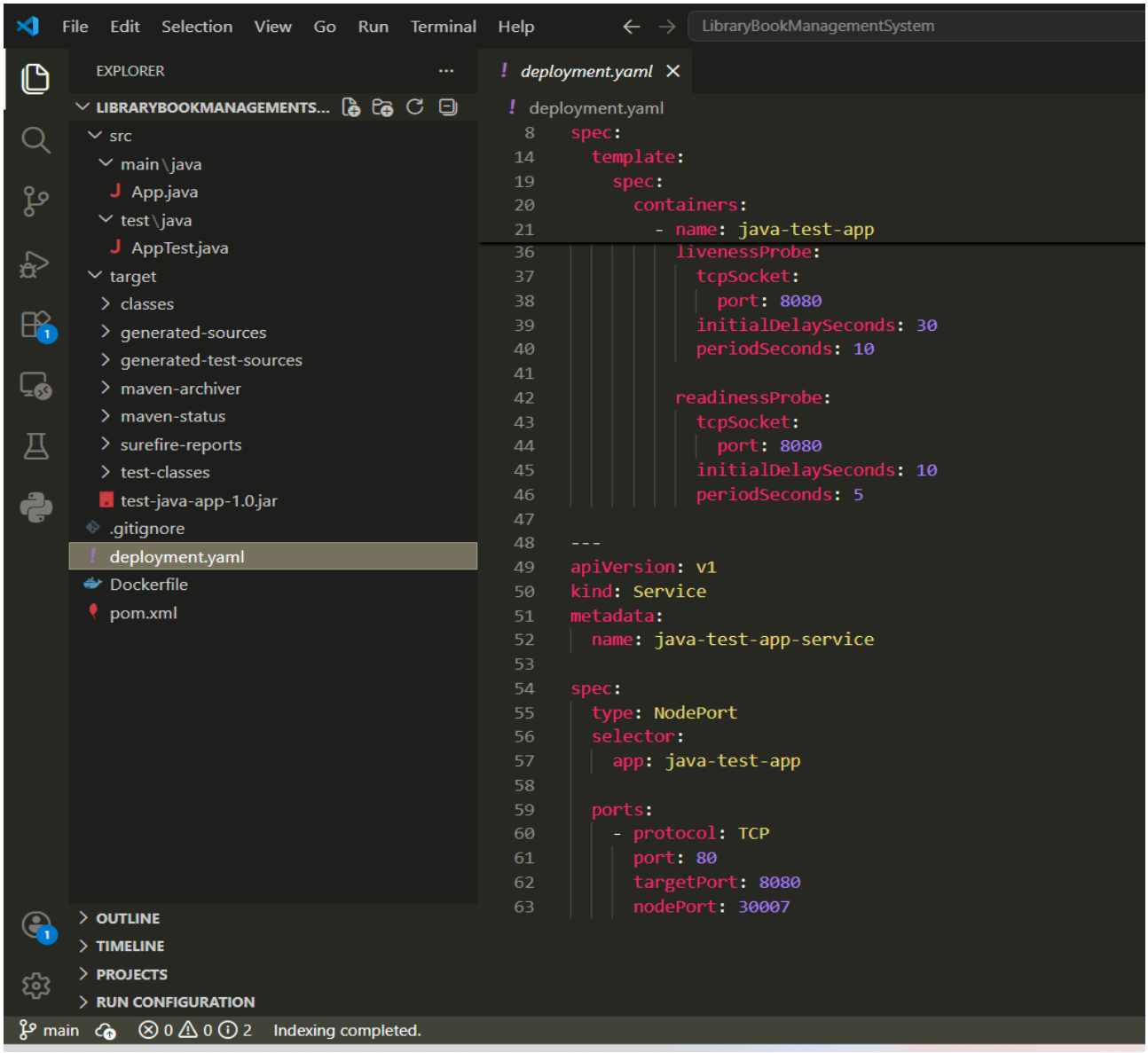
4)Deploy services on Docker And Kubernetes

DockerFile:



Deplyment.yaml:





Deployment.yaml:

File Edit Selection View Go Run Terminal Help LibraryBookManagementSystem

EXPLORER

LIBRARYBOOKMANAGEMENTS...

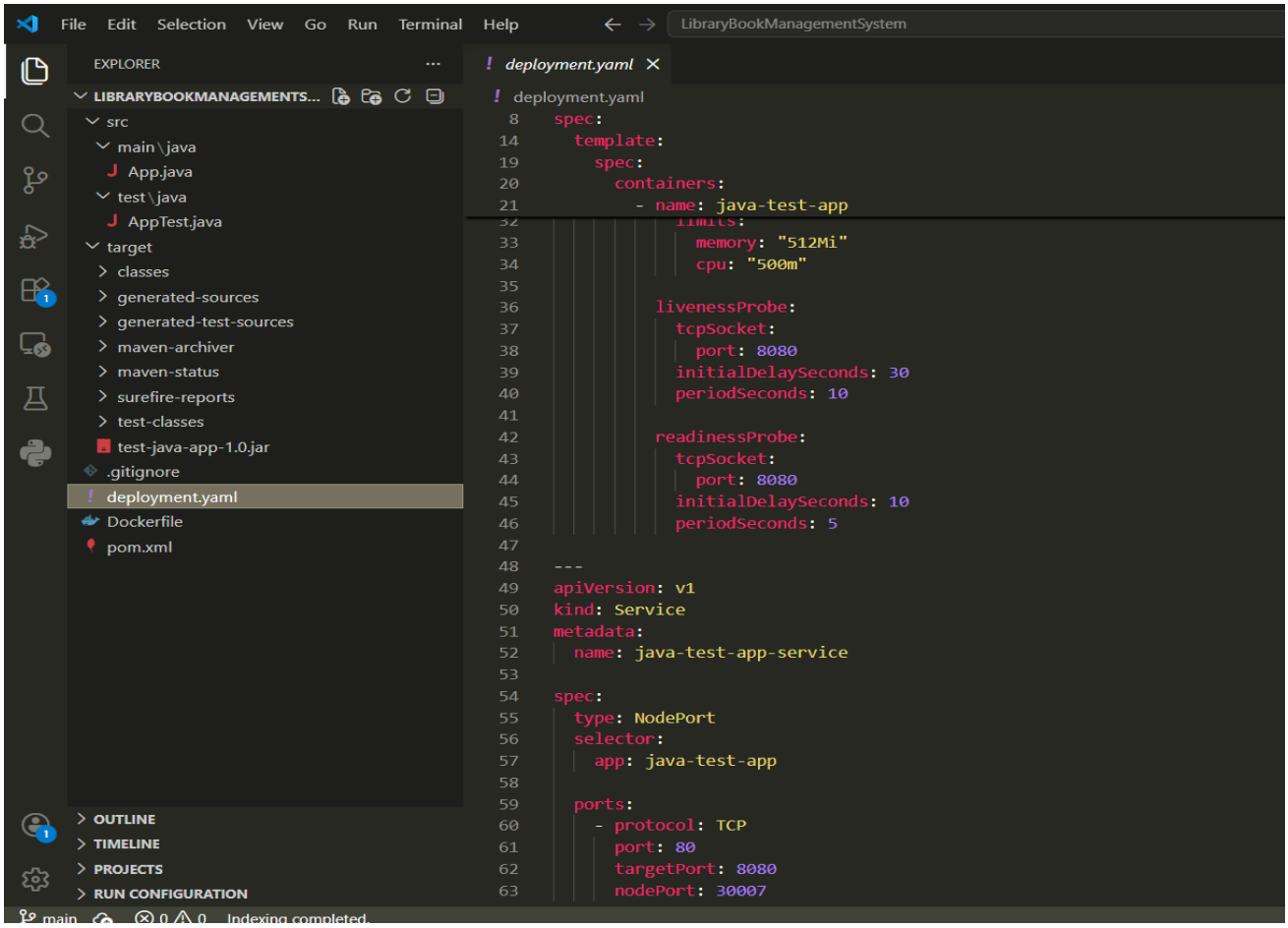
- src
  - main\java
    - App.java
  - test\java
    - AppTest.java
- target
  - classes
  - generated-sources
  - generated-test-sources
  - maven-archiver
  - maven-status
  - surefire-reports
  - test-classes
  - test-java-app-1.0.jar
- .gitignore
- deployment.yaml
- Dockerfile
- pom.xml

OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION

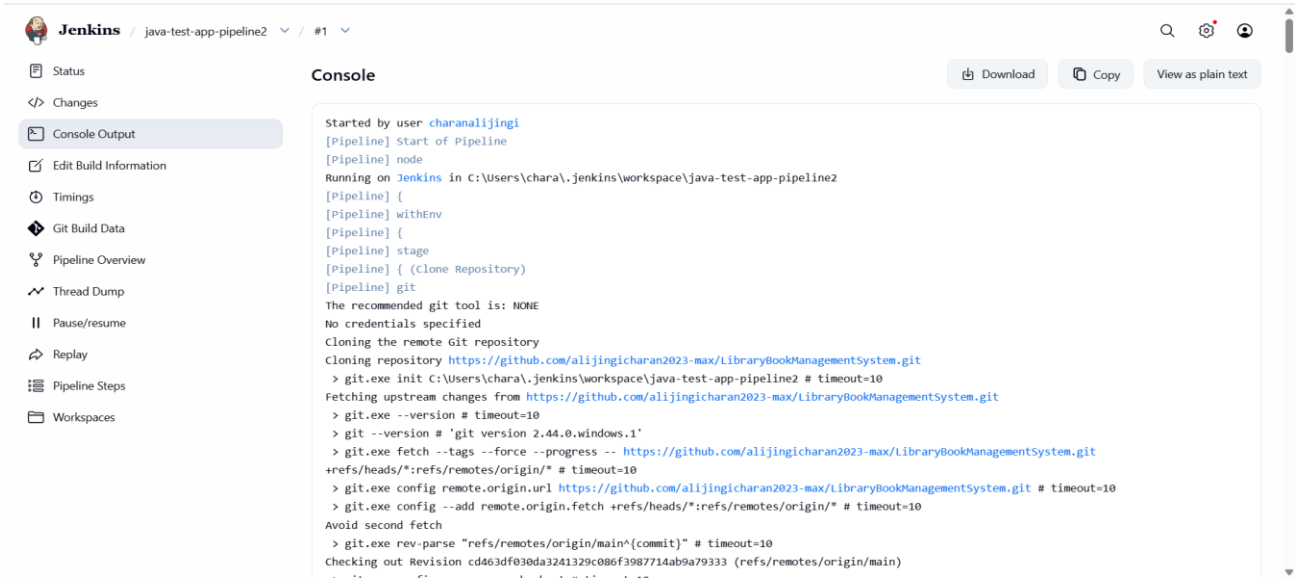
deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-test-app-deployment
  labels:
    app: java-test-app

spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-test-app

  template:
    metadata:
      labels:
        app: java-test-app

    spec:
      containers:
        - name: java-test-app
          image: 23mis0187/java-test-app:latest
          imagePullPolicy: Always

          ports:
            - containerPort: 8080

          resources:
            requests:
              memory: "256Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"

          livenessProbe:
            tcpSocket:
```

main 0 0 Indexing completed.



Console Output:



```
 > git.exe --version # timeout=10
 > git --version # 'git version 2.44.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/alijingicharan2023-max/LibraryBookManagementSystem.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/alijingicharan2023-max/LibraryBookManagementSystem.git # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision cd463df030da3241329c086f3987714ab9a79333 (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f cd463df030da3241329c086f3987714ab9a79333 # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b main cd463df030da3241329c086f3987714ab9a79333 # timeout=10
Commit message: "Removed build files and added gitignore"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build with Maven)
[Pipeline] bat

C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
```

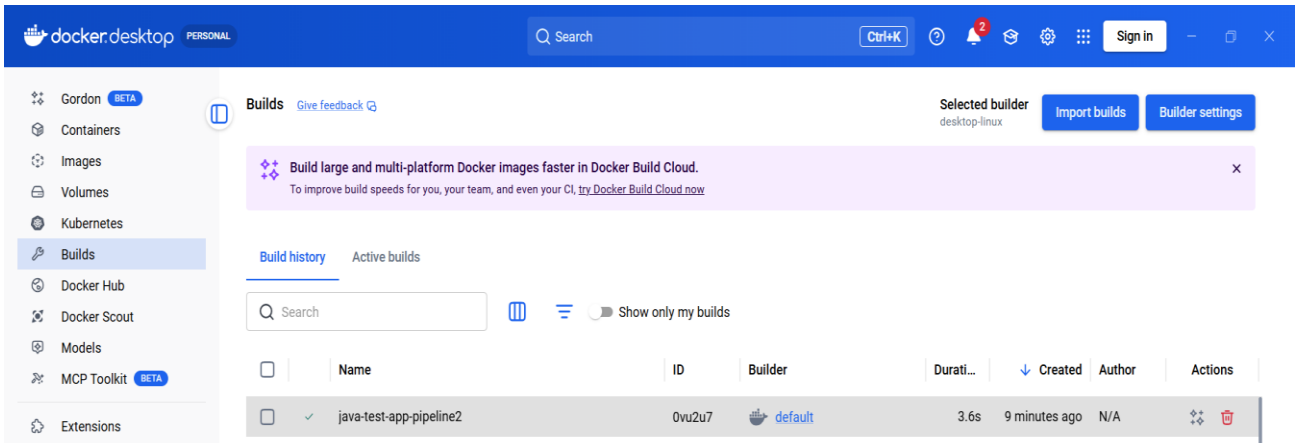
```
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.070 s -- in AppTest
[INFO]
```

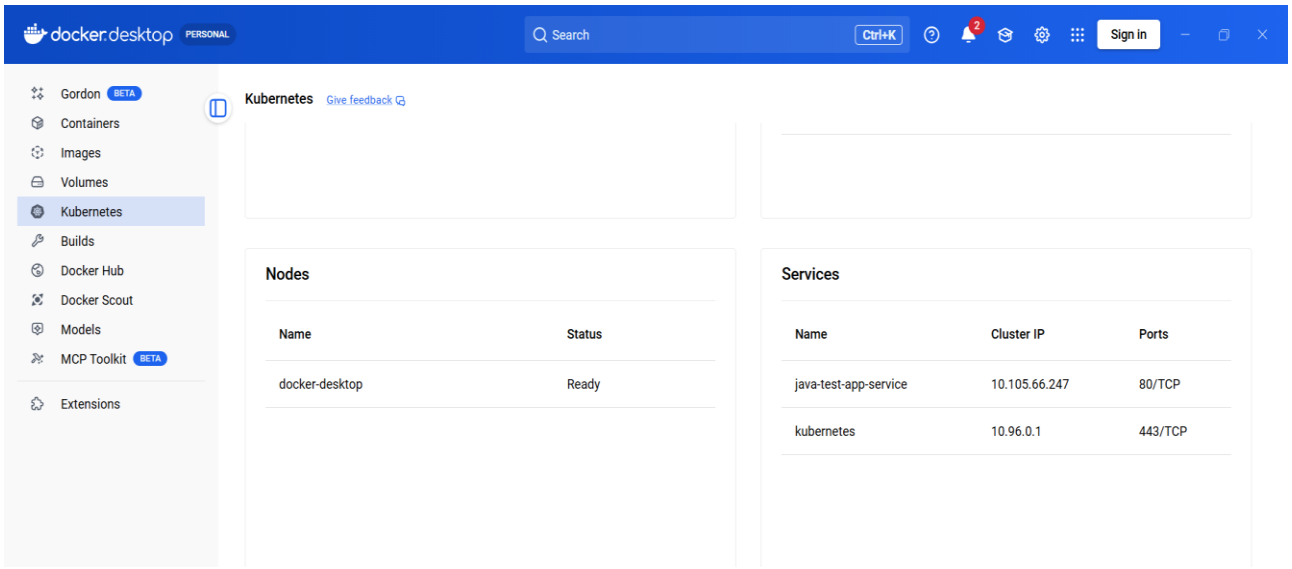
```
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.779 s
[INFO] Finished at: 2026-04-13T10:35:13+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat

C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2>docker build -t 23mis0187/java-app:latest .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 172B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine
#2 ...

#3 [auth] library/eclipse-temurin:pull token for registry-1.docker.io
#3 DONE 0.0s

#2 [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine
```

```
#7 CACHED

#8 [3/3] COPY target/*.jar app.jar
#8 DONE 0.0s

#9 exporting to image
#9 exporting layers 0.1s done
#9 exporting manifest sha256:e1e72fd6c45c2c668481b6feded6423450409bc2cdb43671c7dfb5af3029a700 0.0s done
#9 exporting config sha256:13a3e9dffd869812b203cced3e8d9b72ef2b1cb0607a99d64f6185fdc1a4f444 0.0s done
#9 exporting attestation manifest sha256:6fec091f125e47cf211929794acfc95835d281c30d8f364ab1bff68a2fd2f804 0.0s done
#9 exporting manifest list sha256:d37b5bc1499405398cbe6ad6ca38c641548ab798e51681335d6a99e5e1d97370 0.0s done
#9 naming to docker.io/23mis0187/java-app:latest
#9 naming to docker.io/23mis0187/java-app:latest done
#9 unpacking to docker.io/23mis0187/java-app:latest 0.0s done
#9 DONE 0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/0vu2u7ykcqy4yu8ofp77b1k19
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push to DockerHub)
[Pipeline] withCredentials
Masking supported pattern matches of %Suribabu@1234%
[Pipeline] {
[Pipeline] bat

C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2>echo    | docker login -u  --password-stdin
error: cannot perform an interactive login from a non-TTY device
```

```
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Waiting
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Waiting
82c540078c28: Layer already exists
589002ba0eae: Waiting
dbc53d626950: Waiting
589002ba0eae: Layer already exists
dbc53d626950: Waiting
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Waiting
dbc53d626950: Layer already exists
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Waiting
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Waiting
fc9711546110: Waiting
2eb285c1c1bd: Waiting
3a547da4fe15: Layer already exists
fc9711546110: Waiting
2eb285c1c1bd: Waiting
fc9711546110: Waiting
2eb285c1c1bd: Waiting
```

```
2eb285c1c1bd: Pushed
latest: digest: sha256:d37b5bc1499405398cbe6ad6ca38c641548ab798e51681335d6a99e5e1d97370 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\Users\chara\.jenkins\workspace\java-test-app-pipeline2>kubectl apply -f deployment.yaml
deployment.apps/java-test-app-deployment configured
service/java-test-app-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```






```
PS C:\Users\chara\OneDrive\Documents\LibraryBookManagementSystem> kubectl get svc
NAME                  TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-test-app-service NodePort    10.105.66.247   <none>        80:30007/TCP   14d
kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        15d
PS C:\Users\chara\OneDrive\Documents\LibraryBookManagementSystem>
```